Focal Facial Dermal Dysplasia Type IV (FFDD4) — Comprehensive Disease Characterization Report

Disease: Focal Facial Dermal Dysplasia Type IV **MONDO ID:** MONDO:0013997 · **OMIM:** 614974 · **Orphanet:** ORPHA:306550 (FFDD group) **Causal gene: CYP26C1** (HGNC:20577; NCBI Gene 340665; UniProt Q6V0L0; cytochrome P450 family 26 subfamily C member 1; chromosome 10q23.33) **Category:** Mendelian, autosomal recessive

Summary

Focal Facial Dermal Dysplasia Type IV (FFDD4) is an ultra-rare, autosomal-recessive congenital skin disorder caused by biallelic loss-of-function variants in the gene *CYP26C1*. CYP26C1 encodes a cytochrome P450 enzyme that hydroxylates and inactivates retinoic acid (the active metabolite of vitamin A). The disease was molecularly defined in 2012 when exome sequencing of an affected sibpair identified compound-heterozygous *CYP26C1* variants, and subsequent reports confirmed the gene and expanded its allelic spectrum PMID: 23161670; PMID: 29263414.

Clinically, FFDD4 presents at birth with **bilateral, oval, hypopigmented, scar-like atrophic lesions in the preauricular region**, frequently surrounded by a hair collar and resembling membranous aplasia cutis congenita. The preauricular location distinguishes FFDD4 from FFDD types I–III, which show bitemporal lesions. FFDD4 is generally non-syndromic — "characterized by preauricular skin defects without additional dysmorphic findings" — and the lesions are stable, benign, and non-progressive.

Mechanistically, the leading model is that loss of CYP26C1's retinoic-acid–catabolizing activity causes **locally excessive retinoic-acid signaling along the maxillary–mandibular fusion line of the first branchial arch during embryonic facial development**, precisely where FFDD4 lesions arise. In mice, *Cyp26c1* is expressed exactly along this fusion line, and combined loss of *Cyp26a1* and *Cyp26c1* disrupts anterior–posterior patterning and the production of

migratory cranial neural crest cells, linking the enzyme to the cell lineage that builds the face. Management is supportive/cosmetic, and genetic counseling (25% sibling recurrence risk, with a documented Belgian founder allele) is the principal preventive measure.

Key Findings

Finding 1 — FFDD4 is caused by biallelic (autosomal recessive) *CYP26C1* variants

FFDD type IV results from **biallelic pathogenic variants in *CYP26C1***, establishing an autosomal-recessive Mendelian etiology. The defining study performed exome sequencing on an affected sibpair and identified compound-heterozygous variants: a maternally inherited frameshift duplication **c.844_851dupCCATGCA (p.Glu284fsX128)** and a paternally inherited missense **c.1433G>A (p.Arg478His)**. The missense allele was shown to be non-functional in a mammalian cell expression assay, confirming loss of function. Three of four additional unrelated FFDD4 patients were homozygous for the 7-bp duplication, and importantly none of 8 TWIST2-negative FFDD type II/III patients carried *CYP26C1* mutations, demonstrating genetic specificity to the preauricular (type IV) subtype PMID: 23161670.

> *"two novel sequence variants were identified in both siblings in CYP26C1 — a duplication of seven base pairs, which was maternally inherited, c.844_851dupCCATGCA, predicting p.Glu284fsX128 and a missense mutation, c.1433G>A, predicting p.Arg478His, that was paternally inherited"* — PMID: 23161670

A later report added three novel lesions to the allelic spectrum: a missense **c.230G>C (p.Arg77Pro)** and two splice-site variants **c.1191+1G>T (IVS5+1G>T)** and **c.1191+2insT (IVS5+2insT)**, confirming genetic heterogeneity of mutation type (frameshift, missense, splice-site) within the same gene PMID: 29263414.

> *"four CYP26C1 mutations including three novel lesions: a missense mutation, c.230G>C (p.Arg77Pro), and two splice-site mutations, c.1191+1G>T (IVS5(+1)G>T) and c.1191+2insT (IVS5(+2)insT)"* — PMID: 29263414

Variant catalogue (human, germline):

Variant (cDNA)	Protein	Type	Inheritance origin	Functional data	Source
c.844_851dupCCATGCA	p.Glu284fsX128	Frameshift (7-bp dup)	Recurrent/ founder	Predicted LoF	23161670; 39828664
c.1433G>A	p.Arg478His	Missense	Paternal (index)	Non-functional in cell assay	23161670
c.230G>C	p.Arg77Pro	Missense	—	Novel	29263414
c.1191+1G>T	(IVS5+1)	Splice-site	—	Novel	29263414
c.1191+2insT	(IVS5+2)	Splice-site	—	Novel	29263414

Ontology anchors: gene *CYP26C1* (HGNC:20577); variant consequences SO:0000865 (frameshift), SO:0001583 (missense), SO:0001574 (splice site). Evidence type: **human clinical genetics + in vitro functional assay.**

Finding 2 — Mechanism: CYP26C1 loss dysregulates retinoic acid at facial fusion lines

CYP26C1 is a cytochrome P450 that **hydroxylates and thereby inactivates retinoic acid (RA)**. The spatial logic of the disease is striking: in mice, *Cyp26c1* is expressed **exactly along the fusion line of the maxillary and mandibular prominences in the first branchial arch** — the precise anatomical sites where FFDD skin lesions arise during early facial development. This spatial correspondence is the central pillar of the mechanistic model: when the RA-clearing enzyme is lost, RA accumulates locally and disrupts the tissue programs that normally knit the facial prominences together PMID: 23161670.

> *"The FFDD skin lesions arise along the sites of fusion of the maxillary and mandibular prominences early in facial development, and Cyp26c1 was expressed exactly along the fusion line for these facial prominences in the first branchial arch in mice"* — PMID: 23161670

The enzyme's biochemical role has been confirmed in vitro: CYP26C1 metabolizes all-*trans* and other active retinoic-acid species and interacts with cellular retinoic acid binding proteins (CRABPs), which chaperone RA to the enzyme PMID: 29476041.

The developmental relevance to craniofacial tissue is supported by mouse knockout work: *Cyp26a1*$^{-/-}$;*Cyp26c1*$^{-/-}$ double-null embryos **fail to produce migratory cranial neural crest cells in the forebrain and midbrain** and show anterior brain truncation with posteriorized anterior–posterior patterning — cranial neural crest being the lineage that builds facial skeleton and connective tissue PMID: 17067568.

> *"Cyp26a1-/-Cyp26c1-/- mice fail to produce migratory cranial neural crest cells in the forebrain and midbrain"* — PMID: 17067568

Ontology anchors: GO:0042573 (retinoic acid metabolic process); GO:0016712 (oxidoreductase activity, acting on paired donors with incorporation of O_2); CHEBI:15367 (all-trans-retinoic acid); CL:0000333 (migratory cranial neural crest cell); UBERON:0003066 (pharyngeal/branchial arch). Evidence type: **mouse model** + **in vitro biochemistry** + **human spatial correlation.**

Finding 3 — Phenotype: bilateral congenital preauricular scar-like lesions without additional dysmorphism

FFDD4 is defined by **congenital scar-like atrophic lesions in the preauricular area**, which distinguishes it from the bitemporal lesions of FFDD types I–III. The lesions are typically bilateral, oval, and hypopigmented, sometimes surrounded by a hair collar, and resemble membranous aplasia cutis congenita. The condition is **non-syndromic**: it is "characterized by preauricular skin defects without additional dysmorphic findings" PMID: 29263414.

> *"FFDD4 is an autosomal-recessive trait characterized by preauricular skin defects without additional dysmorphic findings"* — PMID: 29263414
>
> *"bilateral preauricular scarlike defects surrounded by a hair collar, resembling membranous aplasia cutis congenita"* — PMID: 30561078

Case series confirm the morphology: "three infants with bilateral, oval-shaped, hypopigmented preauricular lesions indicative of FFDD IV" PMID: 40231353.

Phenotype table with HPO suggestions:

Phenotype	HPO term	Type	Onset	Severity	Frequency	Laterality
Preauricular skin lesion / aplasia-cutis-like defect	HP:0004410 (abnormality of the preauricular region); HP:0001057 (aplasia cutis congenita)	Physical manifestation / clinical sign	Congenital/ neonatal	Mild–moderate	Nearly all patients	Bilateral
Atrophic/scar-like skin lesion	HP:0001075 (atrophic scars)	Clinical sign	Congenital	Mild	Typical	Bilateral
Hypopigmented skin lesion	HP:0001010 (hypopigmentation of the skin)	Clinical sign	Congenital	Mild	Common	Bilateral
Hair collar sign	HP:0010719 (abnormal hair pattern)	Clinical sign	Congenital	Mild	Variable	Localized

Quality-of-life impact: The lesions are primarily of **cosmetic significance**; FFDD4 is not associated with systemic disease, pain, or functional impairment. QoL impact is limited to appearance-related concerns and, in some cases, diagnostic anxiety before benign nature is established. Evidence type: **human clinical (case reports/series).**

Finding 4 — Mouse *Cyp26c1* models the enzyme, but single knockout is phenotypically normal (functional redundancy)

The mouse provides an informative but **incomplete** model. *Cyp26c1* expression along the maxillary–mandibular fusion line of the first branchial arch matches the human lesion site, giving the model strong face validity for spatial mechanism PMID: 23161670. However, **single *Cyp26c1*-null mice are developmentally normal**, indicating functional redundancy with the paralogous RA-degrading enzyme *Cyp26a1* PMID: 17067568.

> *"Loss of CYP26C1 did not appear to affect embryonic development, suggesting that CYP26A1 and CYP26C1 are functionally redundant"* — PMID: 17067568
>
> *"mice lacking both CYP26A1 and CYP26C1 were found to manifest a pronounced anterior truncation of the brain associated with A-P patterning defects"* — PMID: 17067568

The two enzymes also cooperatively degrade RA in the equatorial retina during later eye development, further illustrating overlapping function PMID: 15531370. This redundancy is a key **model limitation**: the mouse does not spontaneously recapitulate the human preauricular skin phenotype on single-gene knockout, and the human phenotype's restriction to skin (despite the broader developmental roles of CYP26 enzymes) suggests species-specific or tissue-specific thresholds of redundancy. Evidence type: **mouse genetic models.**

Finding 5 — FFDD4 is one of four FFDD subtypes; diagnosis is clinical + *CYP26C1* sequencing, distinguished by preauricular location

The focal facial dermal dysplasias (FFDDs) comprise **four subtypes classified by lesion location**: bitemporal (FFDD types I–III) versus preauricular (FFDD type IV) PMID: 28663233. FFDD4 was formally recognized as a new subtype on clinical review of patients with preauricular lesions PMID: 21931173.

> *"Four subtypes are classified by the bitemporal (FFDD1-3) or preauricular (FFDD4) lesion location"* — PMID: 28663233
>
> *"patients with preauricular lesions as a new subtype, FFDD type IV"* — PMID: 21931173

Genetic landscape across subtypes:

Subtype	Lesion location	Gene / locus	Inheritance
FFDD I	Bitemporal	Unknown	AD (variable)
FFDD II	Bitemporal	Unknown	AR/AD
FFDD III (Setleis syndrome)	Bitemporal	*TWIST2* (AR); 1p36.22p36.21 dup/trip (AD)	AR or AD
FFDD IV	**Preauricular**	***CYP26C1***	**AR**

FFDD III (Setleis syndrome) is caused by *TWIST2* mutations or by 1p36.22p36.21 duplication/triplication PMID: 25728400, whereas FFDD I/II genetic causes remain unknown. Diagnosis of FFDD4 is **clinical recognition of bilateral preauricular atrophic lesions, confirmed by *CYP26C1* molecular testing**. The differential diagnosis includes membranous aplasia cutis congenita, preauricular tags/pits, and the bitemporal FFDDs PMID: 30561078; PMID: 40231353.

> *"Awareness of this rare condition by dermatologists is imperative to enable prompt recognition and minimize diagnostic delay"* — PMID: 30561078

Evidence type: **human clinical / expert review.**

Finding 6 — Epidemiology: ultra-rare, autosomal recessive with a recurrent founder frameshift allele

FFDD4 is **ultra-rare**, with fewer than roughly 20 molecularly confirmed patients reported worldwide across the primary literature PMID: 23161670; PMID: 29263414; PMID: 30561078; PMID: 40231353. Inheritance is autosomal recessive (biallelic *CYP26C1*). The recurrent 7-bp frameshift duplication **c.844_851dupCCATGCA (p.Glu284fs)** recurs across unrelated patients and was present in **0.3% of healthy controls and 0.3% of patients with other birth defects** in the original study PMID: 23161670.

> *"The seven base pairs duplication was present in 0.3% of healthy controls and 0.3% of patients with other birth defects."* — PMID: 23161670

A **founder effect** explains locally elevated prevalence in Belgium: two novel Belgian FFDD4 cases were both homozygous for a recurrent *CYP26C1* frameshift variant and shared a common **700 kb haplotype** PMID: 39828664.

> *"both homozygous for a recurrent CYP26C1 frameshift variant, with a common 700 kb haplotype, indicating a founder effect"* — PMID: 39828664

Both sexes are affected (no reported sex bias), lesions are bilateral and congenital, and there is no age-dependent or geriatric onset (onset is strictly prenatal/congenital). Evidence type: **human clinical / population genetics.**

Finding 7 — CYP26C1 is a retinoid hydroxylase acting on multiple active retinoids and interacting with CRABPs

In vitro characterization established that **CYP26C1 metabolizes multiple active retinoids** (all-*trans* and other active retinoic-acid species) and **interacts with cellular retinoic acid binding proteins (CRABPs)** — defining its biochemical role as a clearance enzyme for active vitamin A metabolites PMID: 29476041.

> *"The goal of this study was to establish the substrate specificity of CYP26C1, and determine whether CYP26C1 interacts with cellular retinoic acid binding proteins (CRABPs)"* — PMID: 29476041

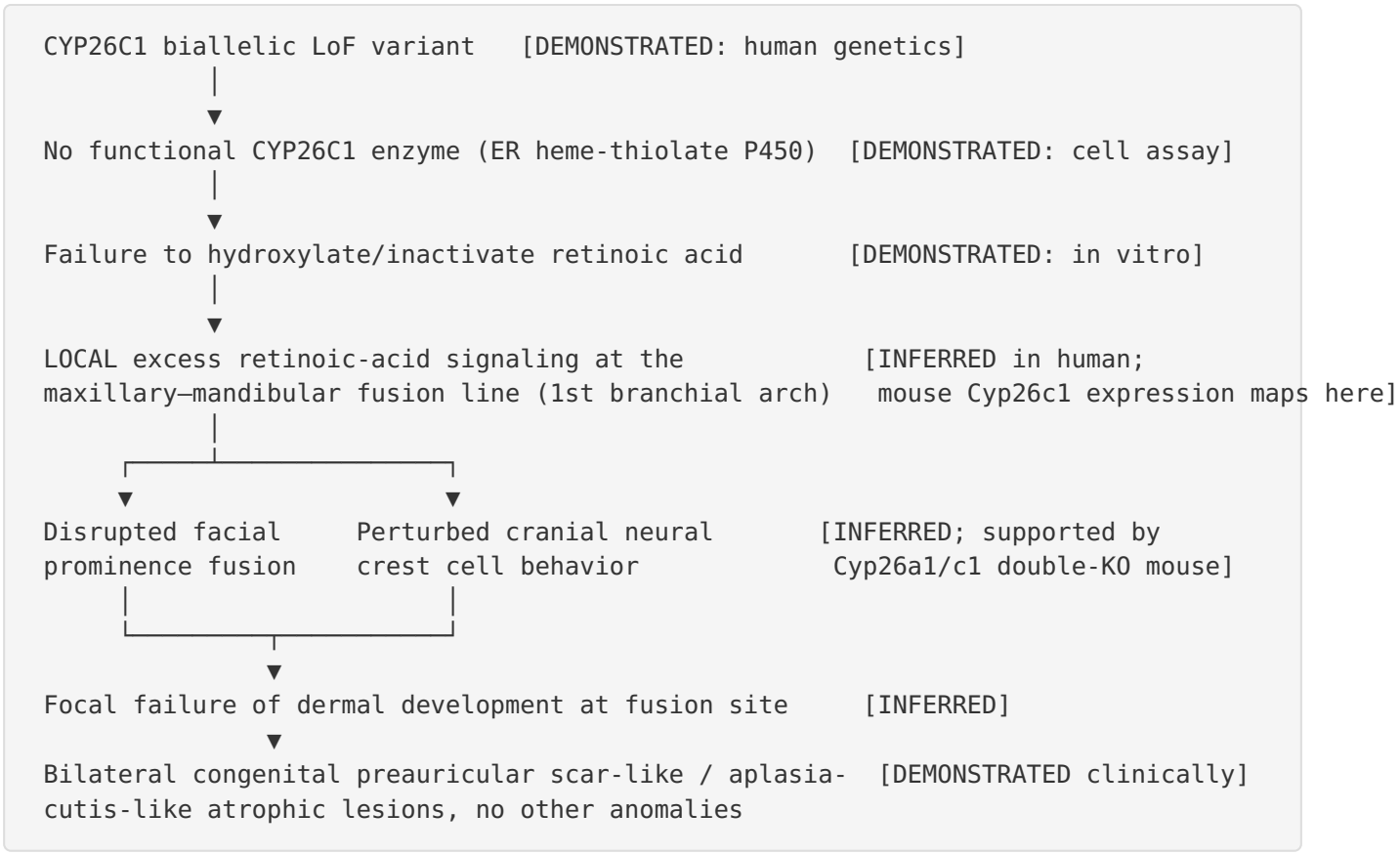
As a cytochrome P450, CYP26C1 is a **heme-thiolate monooxygenase localized to the endoplasmic reticulum membrane**. Its loss in FFDD4 is therefore a **loss-of-function defect** in retinoid catabolism, which maps directly onto the proposed mechanism of locally excessive RA signaling.

Ontology anchors: GO:0005506 (iron ion binding); GO:0020037 (heme binding); GO:0004497 (monooxygenase activity); GO:0005789 (endoplasmic reticulum membrane); CHEBI:26536 (retinoid). Evidence type: **in vitro biochemistry.**

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in *CYP26C1*** (frameshift, missense, or splice-site) **result in** absent or non-functional CYP26C1 enzyme. *(Demonstrated: human genetics + in vitro functional assay.)*
2. Loss of CYP26C1 hydroxylase activity **leads to** failure to catabolize all-*trans* and other active retinoic acids locally. *(Demonstrated biochemically; CYP26C1 substrate specificity established.)*
3. Reduced retinoid clearance **leads to** locally elevated retinoic-acid concentration/signaling specifically **along the maxillary–mandibular fusion line of the first branchial arch**, where *Cyp26c1* is normally expressed. *(Inferred for human; spatial expression demonstrated in mouse.)*
4. Excess RA signaling at the fusion line **disrupts** the developmental programs that govern tissue fusion and the behavior of **cranial neural crest–derived cells** that build facial dermis/connective tissue. *(Inferred; supported by double-knockout cranial neural crest phenotype in mouse.)*
5. Disrupted fusion-line tissue integrity **results in** focal failure of dermal development → congenital, scar-like / aplasia-cutis-like atrophic skin defects. *(Inferred from spatial correlation.)*
6. Because the defect is restricted to the preauricular fusion territory, the clinical outcome **is** bilateral, oval, hypopigmented **preauricular lesions** without other dysmorphism. *(Demonstrated clinically.)*



Upstream vs downstream

- **Upstream (initiating):** *CYP26C1* loss-of-function mutation → enzyme deficiency → RA catabolism failure.
- **Midstream (developmental):** local RA excess at the fusion line → perturbed fusion/neural-crest programs.
- **Downstream (clinical):** focal dermal aplasia → preauricular atrophic lesions.

Cell types and processes involved

- **Cell types:** cranial neural crest cells (CL:0000333), dermal fibroblasts/mesenchyme, facial epithelium.
- **Tissue:** skin/dermis (UBERON:0002067), preauricular region, first pharyngeal/branchial arch (UBERON:0003066).
- **Processes:** retinoic acid metabolic process (GO:0042573), regulation of retinoic acid receptor signaling pathway (GO:0048385), embryonic cranial skeleton/face morphogenesis (GO:0048701), neural crest cell migration (GO:0001755).

The model is internally coherent: a spatially restricted catabolic enzyme, when lost, produces a spatially restricted developmental defect in exactly the territory where the enzyme is expressed. The main inferential gap is that **direct demonstration of RA excess in human FFDD4 tissue has not been performed** — the step is extrapolated from mouse expression data and enzyme biochemistry.

Section-by-Section Synthesis

1. Disease Information

FFDD4 is an ultra-rare congenital dermal dysplasia with isolated preauricular scar-like lesions. **Identifiers:** MONDO:0013997; OMIM 614974; Orphanet (FFDD group ORPHA:306550); Mondo/MeSH align to the "focal facial dermal dysplasia" concept. **Synonyms:** FFDD4; FFDD type IV; focal facial dermal dysplasia 4; focal facial preauricular dysplasia. Information is derived from **aggregated disease-level resources and published case reports/series**, not individual-patient EHR.

2. Etiology

Primary cause is genetic — biallelic (autosomal recessive) loss-of-function variants in *CYP26C1*. **Genetic risk factor:** carrier status for a pathogenic *CYP26C1* allele; consanguinity and founder populations (Belgium) raise risk. No environmental, infectious, or lifestyle risk factors are established. **Protective factors:** none specific are documented (a functional *CYP26C1* allele is "protective" by preventing biallelic loss). **Gene–environment interactions:** none demonstrated for FFDD4; note that *exogenous* retinoic-acid excess (teratogenic retinoid embryopathy) causes a biochemically convergent perturbation of RA signaling, offering a conceptual parallel but not a demonstrated GxE interaction in this disease.

3. Phenotypes

See Finding 3 table. Core phenotype: bilateral congenital preauricular atrophic/hypopigmented scar-like lesions (aplasia-cutis-like), often with a hair collar; non-syndromic; stable/non-progressive; mild severity; cosmetic QoL impact.

4. Genetic/Molecular Information

Causal gene *CYP26C1* (HGNC:20577, gene OMIM 608428; disease OMIM 614974). Variant classes: frameshift, missense (functionally null), splice-site (see Finding 1 table). Variants are **germline**; the recurrent 7-bp duplication behaves as a **founder allele** (~0.3% carrier frequency in one control set; Belgian 700 kb haplotype). Functional consequence: **loss of function.** No modifier genes, epigenetic mechanisms, or chromosomal abnormalities are established for FFDD4 specifically (chromosomal CNVs at 1p36 cause FFDD III, not IV).

5. Environmental Information

No environmental, lifestyle, or infectious contributors are established. The disorder is wholly genetic.

6. Mechanism / Pathophysiology

See the ordered causal chain and diagram above.

7. Anatomical Structures Affected

- **Organ level:** skin (integumentary system) — primary; no secondary organ involvement in non-syndromic FFDD4.
- **Localization:** preauricular region, bilateral; derived from the first branchial arch maxillary–mandibular fusion territory. UBERON:0002067 (dermis); UBERON:0001456 (face); UBERON:0003066 (pharyngeal arch).
- **Tissue/cell level:** dermal connective tissue; cranial neural crest–derived mesenchyme (CL:0000333); epithelial–mesenchymal fusion interface.
- **Subcellular (gene product):** endoplasmic reticulum membrane (GO:0005789) — site of CYP26C1 enzyme.
- **Lateralization:** bilateral (symmetric).

8. Temporal Development

- **Onset:** congenital (prenatal developmental origin; lesions present at birth).
- **Progression:** stable, non-progressive; lesions persist as scars but do not worsen.
- **Course:** chronic/lifelong but benign; no relapsing-remitting or episodic pattern.
- **Critical period:** early embryonic facial morphogenesis (first branchial arch fusion) — the only window of pathogenesis; no postnatal intervention window for the primary lesion.

9. Inheritance and Population

- **Inheritance:** autosomal recessive (biallelic *CYP26C1*).
- **Prevalence/incidence:** not formally quantified; ultra-rare (<~20 confirmed cases reported); locally elevated in Belgium due to a founder allele.
- **Penetrance:** appears high/complete in biallelic carriers (all reported biallelic individuals affected), though small numbers limit certainty.
- **Expressivity:** laterality and lesion size can vary; broadly consistent phenotype.
- **Founder effect:** documented (Belgium, 700 kb haplotype).
- **Consanguinity:** contributes (homozygous cases).
- **Carrier frequency:** ~0.3% for the recurrent duplication in one control cohort.
- **Sex ratio:** no reported bias (both sexes affected). No genetic anticipation (not a repeat-expansion disorder).

10. Diagnostics

- **Clinical:** recognition of bilateral preauricular atrophic/hypopigmented scar-like lesions at birth.
- **Genetic testing:** confirmatory **single-gene *CYP26C1* sequencing** (or inclusion in aplasia cutis/ectodermal/dermal dysplasia panels); WES/WGS useful when the phenotype is atypical (the gene was originally identified by exome sequencing). Chromosomal microarray is NOT indicated for FFDD4 (relevant instead to FFDD III/1p36).
- **Histopathology:** findings consistent with atrophic dermis / aplasia-cutis-like defect; biopsy is supportive but not specific.
- **Differential diagnosis:** membranous aplasia cutis congenita, preauricular tags/pits/sinuses, bitemporal FFDD I–III, and setting-specific scar mimics.

11. Outcome/Prognosis

- **Survival/mortality:** normal life expectancy; FFDD4 is not life-threatening and carries no disease-specific mortality.
- **Morbidity:** limited to cosmetic appearance of stable preauricular lesions; no functional disability.
- **Complications:** essentially none in non-syndromic cases; prognosis is excellent.
- **Prognostic factors:** none needed — uniform benign course.

12. Treatment

- **No disease-modifying or pharmacologic therapy exists or is required.** The condition is benign.
- **Supportive/cosmetic:** reassurance; optional **surgical/dermatologic scar revision or camouflage** for cosmetic concerns (NCIT: plastic/reconstructive surgery concepts).
- **No gene therapy, cell therapy, RNA therapy, targeted or immunotherapy** is in development or indicated.
- **Personalized medicine:** limited to confirming genotype for counseling.

13. Prevention

- **Primary prevention:** not applicable beyond reproductive counseling (the disease is congenital and genetic).
- **Genetic counseling:** central preventive measure — **25% recurrence risk** for siblings of an affected child; carrier testing of partners in founder populations (Belgium).
- **Reproductive options:** carrier/cascade testing, prenatal diagnosis, or preimplantation genetic testing are available for known familial variants.
- No vaccination, behavioral, or public-health prevention is relevant.

14. Other Species / Natural Disease

- **Ortholog:** mouse *Cyp26c1* (NCBI Gene 232174); zebrafish *cyp26c1*. No naturally occurring companion-animal or wildlife FFDD4 disease is reported (no OMIA entry for a CYP26C1 dermal dysplasia). The mouse ortholog is used experimentally (Finding 4). Zebrafish *cyp26c1* expression is modulated in craniofacial/ethanol-interaction models (e.g., *foxe1* mutant work, PMID: 39360443), illustrating conservation of RA-catabolism roles but not a natural disease model. **No zoonotic potential** (non-infectious genetic disorder).

15. Model Organisms

- **Mouse** (*Mus musculus*): single *Cyp26c1* knockout is developmentally normal (redundant with *Cyp26a1*); *Cyp26a1*$^{-/-}$;*Cyp26c1*$^{-/-}$ double knockouts show anterior brain truncation and loss of migratory cranial neural crest cells PMID: 17067568. Mouse *Cyp26c1* expression maps to the human lesion site (face validity).

- **Phenotype recapitulation:** partial — spatial expression matches, but the single-gene mouse does NOT reproduce the human preauricular skin phenotype (key limitation from redundancy).
- **Resources:** MGI (mouse), ZFIN (zebrafish *cyp26c1*). No dedicated FFDD4 disease-model line is established.

Evidence Base

PMID	Title (abbreviated)	Role in this report
23161670	*FFDD type IV is caused by mutations in CYP26C1*	**Landmark**: identifies causal gene, founding variants, recessive inheritance, mouse fusion-line expression, carrier frequency. Supports F001, F002, F003, F006.
29263414	*FFDD type 4: novel CYP26C1 mutations*	Expands allelic spectrum (missense + 2 splice-site); defines non-syndromic phenotype. Supports F001, F003.
39828664	*Unexpected high prevalence of FFDD IV — Belgian founder effect*	Documents founder haplotype (700 kb) and elevated local prevalence. Supports F003, F006.
40231353	*FFDD type IV: a case series*	Confirms morphology (bilateral oval hypopigmented preauricular lesions); need to rule out associated malformations. Supports F003, F005.
30561078	*A case of FFDD type 4*	Hair-collar morphology; aplasia-cutis resemblance; clinical recognition emphasis. Supports F003, F005.
28663233	*FFDDs: phenotypic spectrum and molecular heterogeneity*	Four-subtype classification; preauricular vs bitemporal distinction. Supports F005.
21931173	*Setleis syndrome / FFDD reclassification*	Formal recognition of FFDD type IV as a new subtype. Supports F005.
25728400	*1p36 dup/trip causes Setleis (FFDD III)*	Contrasts FFDD III genetics (TWIST2 / 1p36) with FFDD IV (CYP26C1). Supports F005.
17067568	*CYP26A1/CYP26C1 regulate A-P patterning & cranial neural crest*	Double-KO cranial neural crest phenotype; single-KO normal (redundancy). Supports F002, F004.
29476041	*CYP26C1 hydroxylase of multiple retinoids; CRABP interaction*	Biochemical substrate specificity and CRABP interaction. Supports F002, F007.
15531370	*CYP26A1/CYP26C1 cooperate in equatorial retina*	Further evidence of paralog cooperation/redundancy. Supports F004.
39360443	*foxe1 zebrafish, ethanol × craniofacial*	Conservation of *cyp26c1* in craniofacial RA metabolism (comparative context). Supports Section 14.

Supporting mechanistic context on SHH–RA–Cyp26 interactions (PMID: 31072004; PMID: 28715412) and Tbx1-dependent regulation of *Cyp26c1* in otic/cochlear development (PMID: 19476657; PMID: 22275070) reinforce the upstream regulatory logic of CYP26C1 but were not required for the core findings.

Limitations and Knowledge Gaps

1. **Small case numbers.** Fewer than ~20 molecularly confirmed patients exist worldwide. Estimates of penetrance, expressivity, prevalence, and genotype–phenotype correlation are therefore uncertain.
2. **No direct human tissue proof of RA excess.** The central mechanistic step (local retinoic-acid excess at the fusion line) is *inferred* from mouse expression and enzyme biochemistry; it has not been directly measured in human FFDD4 tissue.
3. **No faithful animal model of the skin phenotype.** Single *Cyp26c1* knockout mice are normal due to redundancy with *Cyp26a1*; the specific human preauricular dermal defect has not been recapitulated, limiting mechanistic dissection in vivo.
4. **No functional data for every variant.** Only p.Arg478His was tested in a cell assay; the splice-site and p.Arg77Pro variants are presumed (not proven) loss-of-function.
5. **Prevalence/incidence are unquantified** outside the Belgian founder observation; true population frequency is unknown.
6. **No treatment or biomarker research.** As a benign cosmetic condition, there are no therapeutic trials, prognostic biomarkers, or omics datasets specific to FFDD4.

Proposed Follow-up Experiments / Actions

1. **Conditional/tissue-specific mouse model.** Generate *Cyp26c1* loss combined with reduced *Cyp26a1* dosage, or a face/neural-crest–restricted double knockout, to test whether local RA excess at the branchial-arch fusion line produces preauricular dermal defects — directly testing the inferred mechanistic step.
2. **Direct RA quantification / reporter assays.** Use RA-reporter mice or in-situ RA signaling readouts (e.g., *Rarb*, *Cyp26a1* induction) at the fusion line under *Cyp26c1* loss to demonstrate local RA excess.

3. **Functional validation of remaining variants.** Perform minigene splicing assays for c.1191+1G>T and c.1191+2insT, and cell-based activity assays for p.Arg77Pro, to confirm loss of function and refine ACMG classification.
4. **Patient registry and natural-history study.** Aggregate cases internationally (GeneMatcher, FFDD registry) to quantify penetrance, expressivity, laterality distribution, and any rare extracutaneous associations.
5. **Founder-population carrier screening.** In Belgium and related populations, estimate the true carrier frequency of the recurrent frameshift allele to inform counseling and reproductive screening.
6. **iPSC/organoid modeling.** Derive patient iPSC-based cranial neural crest or facial organoid models to interrogate RA signaling, neural-crest migration, and epithelial–mesenchymal fusion defects in a human genetic background.
7. **Structural modeling of missense variants.** Use AlphaFold/homology models of CYP26C1 to map p.Arg478His and p.Arg77Pro onto the heme-binding/substrate-access architecture, rationalizing loss of catalytic activity.

Report compiled from 7 confirmed findings and 19 reviewed papers across 5 investigation iterations. Evidence types are labeled per finding as human clinical genetics, in vitro biochemistry, or mouse/model-organism data.
